Exercise path

Start with exercises 1 and 2. Then choose exercise 3 or 4 without help. Exercises 5 to 7 are a bank of transfer, investigation or programming tasks: the teacher assigns one. Exercise 8 is revisited later.

Exercise 9.1 · Transform

For every real x, consider the expressions

$$A = e^{x}e^{2-x}, \qquad B = \frac{(e^{2x})^{3}}{e^{x}}, \qquad C = e^{-x}(e^{x}+1).$$

1. Simplify A and B justifying each grouping of exponents.
2. Expand then simplify C. Explain why a sum does not simplify in the same way as a product.

Exercise 9.2 · Equations and inequalities

Consider an equation and an inequality in a real unknown x :

$$e^{2x-1} = e^{x+3} \quad \text{and} \quad e^{-3x} \geq e^{x-2}.$$

1. Solve each one in $\mathbb{R}$, citing the property of the exponential function that allows the comparison of the exponents.
2. When solving the inequality, explain at which step the direction of the inequality changes. Give the solution set as an interval.

Exercise 9.3 · Product and maximum

The function f is defined on [0; 4] by $f(x) = xe^{-x}$.

1. Calculate $f\prime(x)$, then study its sign on the interval.
2. Draw the variation table of f. Deduce its maximum and the x-value where it is attained.
3. Check that the values at the endpoints do not give a larger maximum. Keep the exact values.

Exercise 9.4 · A modelled population

The following two parts are independent.

Part A - A population. A population is modelled by $P(t) = 200e^{0,1t}$ where $t \geq 0$ is the time elapsed in years. $P(t)$ is a theoretical count, which need not be an integer.

1. Determine the initial count, then the multiplier between times t and $t+1$. Hence deduce the exact annual rate and its percentage, rounded to the nearest hundredth.
2. Explain why this model cannot necessarily describe a real population indefinitely.
3. **Part B - Two curves.** In the same coordinate system, sketch the graphs of $t \mapsto e^{\frac{t}{2}}$ and $t \mapsto e^{-\frac{t}{2}}$. Describe how each one varies, their common point and their symmetry. Label each curve clearly.

Exercise 9.5 · Search for a threshold

The amount of a substance, in milligrams, is modelled at time $t \geq 0$ by $Q(t) = 80e^{-\frac{t}{2}}$, and t is expressed in hours. Checks take place at integer times $n = 0, 1, 2, \ldots$. We seek the first check for which the quantity is strictly less than 10 mg.

1. Write an algorithm to find this first integer. Include a bound on the number of iterations and specify what the program returns if the bound is reached.
2. Run the algorithm, then numerically check the quantity at the index found and at the previous index.
3. Justify, using the decrease of Q, that this double check shows that it is indeed the first index. Do not use logarithms.

Exercise 9.6 · A symmetry

Consider the function f defined on $\mathbb{R}$ by $f(x) = e^x + e^{-x}$.

1. Study the parity of f and interpret the result for its curve.
2. Prove that $f(x) \geq 2$ for every real x, choosing a method. State the equality case.
3. Suggest checking the result by a second method, using a square or a study of variations.

Exercise 9.7 · Uniqueness under an assumption

Extension. Assume that the exponential function has already been constructed. Let f a function differentiable on $\mathbb{R}$ which satisfies $f'(x) = f(x)$ for every real x and $f(0) = 1$. Set $g(x) = f(x)e^{-x}$.

1. Calculate g' and explain why g is constant on $\mathbb{R}$.

2. Determine this constant and then deduce from it the expression of f.

3. Does this proof establish the existence or the uniqueness of the required function? Justify.

Exercise 9.8 · Delayed review

Define the sequence (u_n) by $u_n = e^{2n+1}$ for every natural number n.

1. Prove that this sequence is geometric, then give its first term and common ratio.

2. Calculate $u_0 + u_1 + u_2$ in exact form. Compare the direct calculation with using a formula for a sum.

Independent work. Do only the task assigned at the end of the session, as set in the dated scheme of work: 10-20 min maximum. The exercises depend on the concepts already studied; some questions are therefore postponed. Revisit exercise 8 one week after the chapter (10 min). If the planned time is exceeded, note the difficulty and bring your attempts; do not try to do the whole reserve.

To be done in the CQFD textbook

These exercises are done in your textbook. The scheme of work gives the session; they replace the homework planned for that day (10 to 20 minutes in total).

CQFD, exercise 18 p. 187, questions a, b and c. Simplify the products and quotients, citing the property of exponentials used.

CQFD, exercise 30 p. 187, questions a, b and c. Solve the three inequalities and write the solution sets in the form of intervals.